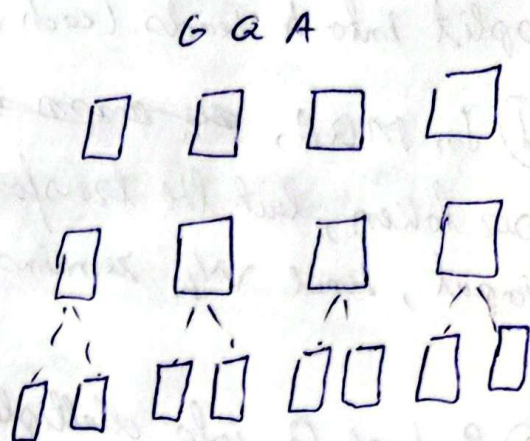
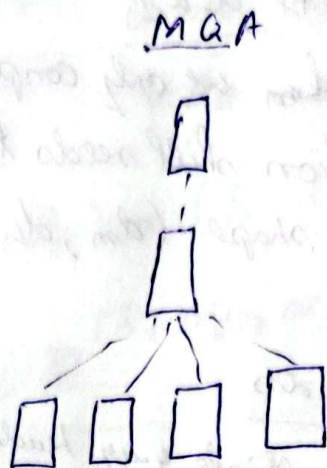
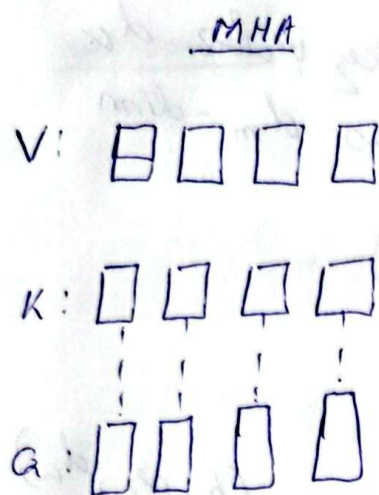


MQA (without KV-caching)



Dimensions

$$X \in \mathbb{R}^{(n_{seq}, d_m)}$$

$$W_Q \in \mathbb{R}^{(d_m, d_m)}$$

$$W_K \in \mathbb{R}^{(d_m, d_k)}$$

$$W_V \in \mathbb{R}^{(d_m, d_k)}$$

n_{seq} = sequence length

d_m = embedding size

h = number of query heads

$d_k = d_{model} / h$; query head dim.

① Compute Q, K, V Projections

$$Q = X^{(n_{seq}, d_m)} \cdot W_Q^{(d_m, d_m)} ; K = X^{(n_{seq}, d_m)} \cdot W_K^{(d_m, d_k)}$$

$$V = X^{(n_{seq}, d_m)} \cdot W_V^{(d_m, d_k)}$$

$$\Rightarrow Q \in \mathbb{R}^{(n_{seq}, d_m)} ; K, V \in \mathbb{R}^{(n_{seq}, d_k)}$$

$\hookrightarrow W_K, W_V$ are diff because MQA has only 1 KV head.

Computational Cost

$$O(n_{seq} \cdot d_m^2) + O(n_{seq} \cdot d_m \cdot d_k) + O(n_{seq} \cdot d_m \cdot d_k)$$

$$\approx \boxed{O(n_{seq} \cdot d_m^2)} \text{ because } d_k = d_m / h$$

Memory Cost

$$\underbrace{O(n_{seq} \cdot d_m)}_Q + \underbrace{O(n_{seq} \cdot d_k)}_K + \underbrace{O(n_{seq} \cdot d_k)}_V \approx \boxed{O(n_{seq} \cdot d_m)}$$

! In standard MHA, V_L maps to (d_m, d_k) and then the output is split into h heads (each of size d_k)

① In MQA, ~~V_L maps to d_m~~ we only compute 1x key vector d_k per token, but the transformation still needs to process d_m -dim input, hence V_L remains of shape (d_m, d_k)

② Reshape Q into Multiple Heads

We reshape Q into multiple query heads:

$$Q^{(n_z, d_m)} \rightarrow Q^{(n_z, h, d_k)} \xrightarrow{\text{transpose}} Q^{(h, n_z, d_k)}$$

final: $Q \in \mathbb{R}^{(h, n_z, d_k)}$

- > no additional computation, just reshaping
- > for memory, we only reshape no extra memory.

③ Process K, V (Single Head)

Unlike MHA, MQA shows K, V across all heads
Instead of reshaping per head, we expand K, V across all heads:

$$K \rightarrow (1, n_z, d_k) \Rightarrow (h, n_z, d_k)$$

$$V \rightarrow (1, n_z, d_k) \Rightarrow (h, n_z, d_k)$$

final shapes:

$$K, V \in \mathbb{R}^{(h, n_z, d_k)}$$

Computational Cost: broadcasting is not a computational cost, so its zero

memory:

$$K, V \text{ are expanded} \Rightarrow (h, n_z, d_k) \Rightarrow \boxed{O(h \cdot n_z \cdot d_k)}$$

④ Compute Attention Scores

we compute:

$$Att_i = \frac{Q \cdot k^T}{\sqrt{d_k}}$$

$$(h, n_2, d_k) \times (h, d_k, n_2) = (h, n_2, n_2)$$

$$A \in \mathbb{R}^{(h, n_2, n_2)}$$

Computational Cost:

→ matrix multiplication complexity

$$O(h \cdot n_2^2 \cdot d_k) \approx O(n_2^2 \cdot d_m)$$

Memory Cost

$$\text{for } A \rightarrow (h, n_2, n_2) \rightarrow O(h \cdot n_2^2)$$

(then we apply softmax, ...)

⑤ Compute output O

$$O = A \cdot V$$

$$(h, n_2, n_2) \times (h, n_2, d_k) = (h, n_2, d_k)$$

$$O \in \mathbb{R}^{(h, n_2, d_k)}$$

Computational Cost

$$O(h \cdot n_2^2 \cdot d_k) \approx O(n_2^2 \cdot d_m)$$

Memory Cost:

$$\rightarrow \text{output } O \rightarrow (h, n_2, d_k) \rightarrow O(h \cdot n_2 \cdot d_k)$$

⑥ Concatenate heads & Project Wo

→ we reshape back to the original shape

$$(h, n_2, d_k) \rightarrow (n_2, h, d_k) \rightarrow (n_2, d_m)$$

final projection:

$$(n_2, d_m) \times (d_m, d_m) = (n_2, d_m)$$

Computational cost:

$$O(n_2 \cdot d_m^2)$$

Memory Cost:

final output $\rightarrow O(n_2 \cdot d_m)$

Conclusion:

Total Computational Cost =

$$O(n_2 \cdot d_m^2) + O(n_2^2 \cdot d_m)$$

Total Memory Cost =

$$O(n_2 \cdot d_m) + O(h \cdot n_2^2)$$

① Key difference MHA reduce memory usage and inference cost by storing K, V across all query heads